KiCad – Training
&
PCB and Material Day

March 2025

Preclass 31.03.25-04.04.25

Stand 28.03.2025

Prof. Dr.-Ing. Stephan Bannwarth, Manuel Leidel

Agenda

- Purpose and Content of the Training
- Training Organization
- Before the training

Purpose and Content of the Training

Purpose

- It is a mandatory preclass for the module “System Driven Hardware Design”
- It is PCB, PCB Materials, Signal Integrity and tool training for KiCad, since you have to develop your own PCB during the master course “System Driven Hardware Design”, which is going to be manufactured
- You have to get familiar with Moodle, especially how to upload files
- We have to test online question and answers with different possibilities

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Content

- What is a PCB and what materials are used?
- How is a PCB manufactured and what constraints does it give to my design?
- What physical properties do PCB materials have?
- What is signal integrity? How do I measure signal integrity? How to layout high speed signals properly?
- How do I use the CAD tool KiCad?
- You are going do enter a new schematic and layout a 2-layer PCB

EYE KNOW HOW
HIGH SPEED SIMULATION AND MEASUREMENT

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Organization at a glance

The course will contain *lectures* and *exercises* for every day, which will be visible in Moodle at the beginning of every day. Lecture material will be videos, documentation or schematics for your own study. With the corresponding exercises you train your gained knowledge and design your own PCB, which you have to upload to Moodle at the end of the trainng. We will review this uploads and have a online meeting on the last days, to discuss your PCBs.

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Type of Work packages

Lecture	Material to gain knowledge, i.e. videos, pdfs, power points, book chapters etc. → Moodle
Exercise	Train your knowledge by your self, depending on the work package you have to hand in results → Moodle
Online Zoom Lecture	Video conference, chat to clarify questions, which arise
On-site attendance	Mandatory to join, attendence will be checked to pass the preclass

Study Time

- Material will be available at the corresponding day in Moodle.
- Your work on your own.
- Question Hours start in time and finish, if there are no further question

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Course Day	Date	Work Package	Type of Work Package	Time / Study Time
Day 1	31.03.25	Introduction to KiCad Training	Online Zoom Lecture	14:00h - 15:30h
		Schematic Editor	Video Lecture	1h
		Schematic Entry	Exercises	2h
Day 2	01.04.25	Layout Editor	Video Lecture	1h
		Layout Entry	Exercise	3h
		Question and Answers on Exercises / KiCad	Online Zoom Lecture	18:00-19:00h
Day 3	02.04.25	Signal Integrity (Mr. Ruckerbauer)	Online Zoom Lecture	14:00h-17:00h
	2.04.25 at 23:59h	Deadline Upload	Upload your KiCad Project	
Day 4	03.04.25	Review of your uploads	C19/0.1	15:00-17:30h
Day 5	04.04.25	PCB Base Materials (Mr. Sommer)	Attendence C19/0.1	09:00h-12:00h
		PCB Basic (Mr. Süßlin)	Attendence C19/0.1	13:30h-16.30h

Before the Training

1. Check if you are enrolled to the Moodle Course for the KiCad Training: *(Please inform your colleagues, if they haven yet signed in, it is mandatory in order to follow "System Driven Hardware Design")*

2. Download and install the latest KiCad version (8.X) on your computer: https://www.kicad.org/

1. Read the KiCad Tutorial: *"Getting Started in KiCad"* (→ Moodle)

2. Get familiar, how you can upload files and finished exercises in Moodle (→ Moodle Documentation)